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Abstract

Aims

In patients with heart failure (HF) influenza vaccination has shown beneficial effects in preventing cardiac decompensations. However, no conclusive results have been achieved in the few studies that have evaluated the impact of vaccination during episodes of acute HF (AHF) decompensation. We conducted a systematic review and meta-analysis to determine the possible effects of influenza vaccination on all-cause mortality in patients diagnosed with AHF.

Methods

PubMed, Medline, Cochrane Central Register of Controlled Trials and Cochrane Database of Systematic Reviews databases were searched for longitudinal studies comparing patients with AHF vaccinated against influenza with unvaccinated patients. The primary outcome selected for meta-analysis was 1-year all-cause mortality, and secondary outcomes consisted of other outcomes reported in at least in two different studies. Statistical heterogeneity was determined by calculating the I^2 statistic. Individual adjusted results were pooled using a random effects model. Sensitivity analysis was run for the primary outcome by removing each individual study and then re-doing the meta-analysis.

Results

Up to 30 June 2025, five observational cohort studies examining the effect of influenza vaccination on 1-year all-cause mortality in AHF patients had been published. Statistical heterogeneity was low ($I^2 = 33.7\%$), meaning that between-study results were consistent. Pooled analysis of confounder-adjusted hazard ratio (HR) for all-cause mortality in vaccinated patients was 0.89 (95% CI 0.83–0.96) compared with unvaccinated patients. All sensitivity analyses rendered very similar results. In-hospital and 90-day mortality were reported in three and two studies and showed similar reductions in risk, with an adjusted odds ratio of 0.85, 95% CI 0.70–1.01, and adjusted HR of 0.86, 95% CI 0.76–0.96; respectively. Isolated data from single studies suggest no effect on hospitalization following discharge after the AHF episode.

Conclusions

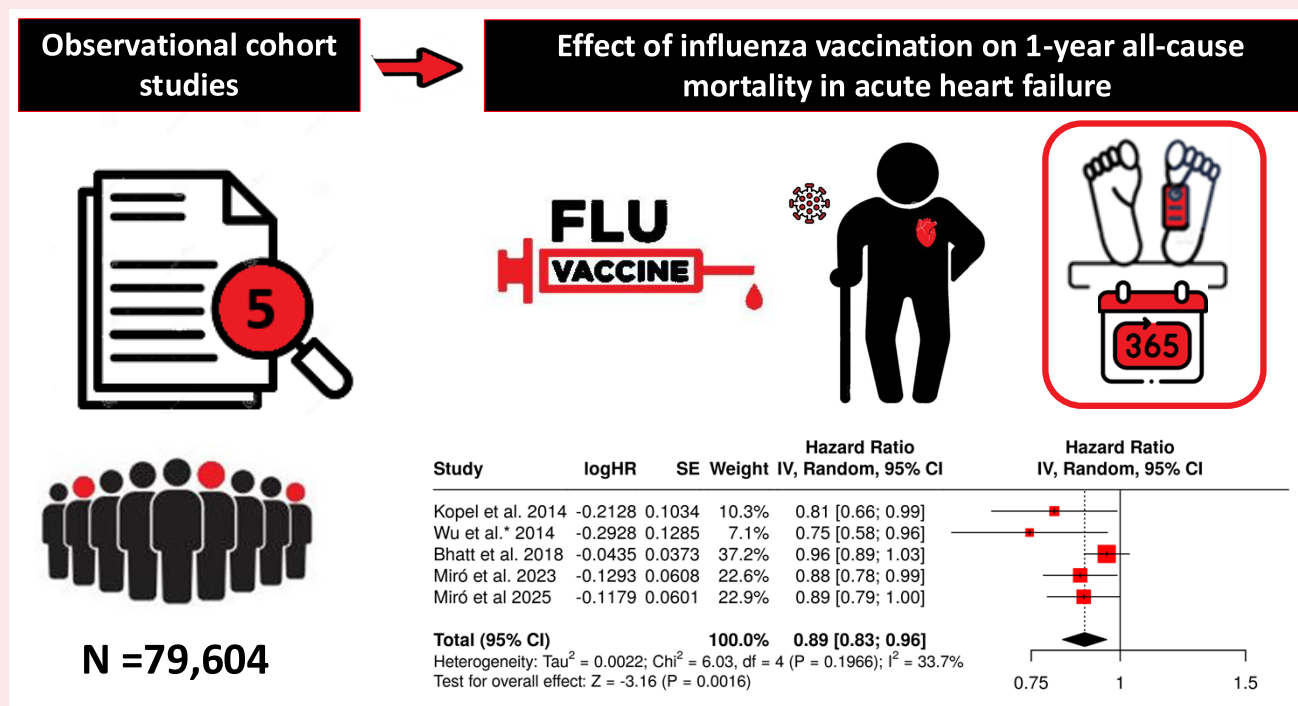
Influenza vaccination is associated with a lower short- and long-term all-cause mortality in patients with decompensated HF; however, as all the studies included in this meta-analysis were observational, these results could be subject to residual confounding and causality cannot be directly inferred from them.

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Graphical abstract



Influenza vaccination is associated with a lower short- and long-term all-cause mortality in patients with decompensated heart failure.

Keywords

Influenza • Influenza vaccination • Heart failure • Acute heart failure • Decompensated heart failure • Mortality

Introduction

Heart failure (HF) is a chronic cardiovascular condition with a high prevalence among individuals over the age of 65.¹ Both its incidence and prevalence are increasing, likely as a result of the rising life expectancy and ageing populations.² Acute decompensations of HF—referred to as acute heart failure (AHF)—are associated with considerable morbidity and mortality, with in-hospital death rates ranging between 5% and 10% across large-scale registries.^{2–6} Prompt identification and management of factors contributing to these decompensations are essential to improving outcomes. In addition to in-hospital mortality, AHF episodes mark the onset of a well-documented ‘vulnerable phase’ that prolongs for several months after discharge.^{7–9} During this period, patients face an elevated risk of rehospitalization and death. One-year all-cause mortality after an AHF episode has been estimated at ~30%,^{2,5,6} underscoring the need for preventive strategies that extend beyond the acute phase.

Among the precipitating factors of AHF, respiratory infections—particularly influenza—play a major role.^{10–12} Influenza vaccination is known to reduce the incidence and severity of such infections,¹³ and multiple studies have reported its protective effect on mortality and cardiovascular events in patients with chronic HF.^{14–17} Patients with HF frequently present with multiple comorbidities, frailty, and functional dependence,^{2,5,9,18,19} further justifying the recommendation of influenza vaccination in this population to prevent decompensation events.

However, the extent to which influenza vaccination influences outcomes in patients already experiencing an episode of AHF, regardless of its trigger, remains less well explored. In previous studies by our group, preliminary observational data suggest a possible benefit even during acute decompensations requiring emergency department (ED)

care, although limited sample sizes and heterogeneous outcomes have precluded firm conclusions.^{20,21} Therefore, given the potential clinical impact of influenza vaccination in this high-risk population—for both primary and secondary prevention—,²² we conducted a systematic review and meta-analysis of the existing literature to assess the effect of influenza vaccination on clinical outcomes specifically in patients with AHF.

Methods

Study selection and data extraction

This systematic review was conducted and reported using PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)²³ and meta-analysis of observational studies in epidemiology²⁴ guidelines as the reference. It was uploaded in the PROSPERO registry with the reference number CRD42022313648.

Longitudinal studies evaluating the effect of influenza vaccination on patients with AHF published up to 30 June 2025 were identified using PubMed, Medline, Cochrane Central Register of Controlled Trials, and the Cochrane Database of Systematic Reviews databases. The following key words and Medical Subject Headings were used: ‘influenza vaccine’ and ‘heart failure’ and ‘mortality’. We limited our literature search to studies involving human subjects, with a sample size greater than 1000 individuals and to those published in English. Additionally, we manually searched the references that were cited in the relevant publications. Inclusion criteria for the present meta-analysis was the assessment of influenza vaccination in patients with AHF, with information about outcomes including at least data on mortality. Studies had to include a vaccinated group and a comparison group in which patients had not been vaccinated against influenza. Studies analysing patients with stable, non-decompensated HF were excluded from this systematic review and meta-analysis, as our main objective

was to establish the influence of vaccination on outcomes of HF patients with an acute decompensation.

Citations were uploaded to the reference manager software Mendeley Desktop (version v2.136.0) for title and abstract screening and data characterization. Titles and abstracts were reviewed by two independent reviewers (N.I.-O. and P.L.) and potentially eligible reports were retrieved and the full text scrutinized for inclusion. A third investigator or 'tie-breaker' (O.M.) was involved in the case of disagreement. Data extraction included: first author name, study period and year of publication, country in which the study was conducted, source of data, study design, number of patients included, age, gender, and adjusted primary (1-year all-cause mortality) and secondary outcomes (all the remaining reported outcomes). Some remarks considered relevant for the data analysis and discussion were also recorded and are shown in Table 1, together with the above commented variables.

Outcomes

One-year mortality was the primary outcome for the meta-analysis. For publications that did not report this primary outcome, the corresponding authors were contacted and invited to provide the adjusted ratios with the 95% confidence intervals (CI). The remaining outcomes reported in any of the studies included were considered secondary outcomes.

Statistical analysis

Individual results for the primary outcome (1-year all-cause mortality) and for the secondary outcomes reported in more than one study were pooled for meta-analysis. The adjusted odds ratios (ORs) or hazard ratios (HRs) of the studies were combined using a random effects meta-analysis, which assumes that the included studies represent a random sample from a larger universe of potential studies. Estimations were made with a 95% CI. Statistical heterogeneity was determined by calculating the I^2 statistic, which measures the percentage of variance across studies due to heterogeneity rather than chance.

For the primary outcome, the potential for publication bias was investigated using visual assessment of the funnel plot. We also ran a sensitivity analysis by excluding each study one by one to identify those with a substantial impact on the overall effect size. After individual study eliminations from the meta-analysis, we recalculated the pooled effect with the remaining studies. Finally, a cumulative meta-analysis was also performed by sequentially adding studies in chronological order to assess how the overall effect size evolved as evidence accumulated over time.

All the statistical analyses were performed with MetaAnalysisOnline.com software.²⁵

Results

The article selection process is shown in Figure 1. The initial search yielded 149 articles. After removing duplicates and screening of the title and abstract, 15 studies remained for potential inclusion. However, after complete reading and analysis of data contained in these studies, only 5 were finally included for meta-analysis. There were no exclusions due to the study sample being less than 1000 individuals. The five studies included were non-intervention studies, and their main characteristics are summarized in Table 1. They involved a total of 79 604 patients with AHF. The mean or median age of the patients in these studies ranged between 70 and 85 years. There were two studies conducted in the USA in 2014²⁶ and in 2018²⁷ and the remaining three were from Europe: one in Israel in 2014^{28,29} and two in Spain in 2023²⁰ and 2025.²¹ The median age of the populations ranged from 70 to 85 years, with the proportion of male participants varying notably between studies—from 98.2% in the study by Wu et al.²⁶ based on a predominantly male veteran population, to 43.4% in that of Miró et al.²¹

The number of vaccinated vs unvaccinated patients differed across studies, with the lowest vaccination rate in the Miró et al. cohort²⁰ (18.5%, 1139 vaccinated vs 5008 unvaccinated) and the highest in the cohort of Wu et al.²⁶ (78.8%, 2087 vaccinated vs 429 unvaccinated,

which decreased to 76.2% after omitting 223 patients that refused vaccination from the analysis. Remarkably, no study reported the vaccine type, the specific influenza strain administered, or whether the vaccine matched the circulating seasonal strains. Vaccination timing corresponded to the seasonal campaign in all the studies; additionally, in the study by Wu et al.²⁶ patients vaccinated during the index AHF hospitalization were also classified as vaccinated, although the number vaccinated in-hospital was not reported. All the studies enrolled patients hospitalized for an AHF episode; while Miró et al.²⁰ and²¹) also included patients diagnosed with AHF in the ED who were discharged directly home. In two studies—Wu et al.²⁶ and Bhatt et al.²⁷—mortality analyses were restricted to patients discharged alive, excluding in-hospital deaths.

The five studies included reported data on mortality, and four reported the 1-year all-cause mortality: in the study by Wu et al.²⁶ the 1-year mortality was reported as an adjusted OR, while the studies of Kopel et al.,²⁸ Bhatt et al.,²⁷ and Miró et al.²¹ reported HRs. The remaining study only reported adjusted in-hospital and 90-day mortality. We contacted the authors to ask about the HR for 1-year mortality adjusted for the same covariates as the rest of the outcomes reported in the original study and they stated that the 1-year all-cause mortality for vaccinated patients was 0.878 (95% CI 0.775–0.994).

Figure 2 illustrates the results of the meta-analysis assessing the association between influenza vaccination and 1-year all-cause mortality among patients with AHF. The pooled analysis, based on five cohort studies, showed a statistically significant reduction in the risk of death among vaccinated patients (Panel A). The overall ratio was 0.89 (95% CI 0.83–0.96), indicating a relative reduction of 11% in all-cause mortality associated with influenza vaccination. The analysis was conducted using a random effects model with inverse variance weighting, and the test for overall effect was significant at $P < .05$. Importantly, heterogeneity across studies was negligible (I^2 33.7%), indicating that the effect sizes across studies were uniform in both magnitude and direction, which suggests that the protective effect of influenza vaccination observed was consistent. Sensitivity analyses (in which each study was sequentially omitted to assess its individual impact) confirmed the robustness of the results: the pooled HRs remained stable and statistically significant across all scenarios, with the HR ranging between 0.86 and 0.91 (Panel B). Of note, even after removing the Wu et al. study,²⁶ which reported 1-year risk as an OR instead of a HR (as in the four remaining studies), the pooled adjusted HR still remained statistically significant (0.91, 95% CI 0.86–0.97). This consistency reinforces the reliability of our results showing an association of influenza vaccination with a lower 1-year mortality in AHF patients. The cumulative meta-analysis showed a trend of a decreasing effect of influenza vaccination on 1-year mortality over time (Panel C).

To explore potential publication bias for the primary outcome (1-year mortality), a funnel plot was constructed (Figure 3). Although formal testing, such as Egger's test, was not performed due to the limited number of studies included, visual inspection of the funnel plot did not reveal substantial asymmetry. This suggests a low risk of publication bias influencing the association observed between influenza vaccination and 1-year all-cause mortality in patients with AHF. Nonetheless, the small number of studies available limits the conclusiveness of this analysis.

Figure 4 presents the pooled effect estimates for the secondary outcomes reported in at least two studies: in-hospital all-cause mortality (reported in three studies; Panel A) and 90-day all-cause mortality (reported in two studies; Panel B). The Kopel et al. study²⁸ reported in-hospital mortality limited to a maximum of 30 days, and was expressed as a HR. Meta-analysis showed a pooled adjusted OR of 0.84 (95% CI: 0.70–1.01) for in-hospital mortality and an adjusted HR of 0.86 (95% CI: 0.78–0.95) for 90-day mortality, indicating a relative reduction in mortality of 16% and 14% during admission and the early post-discharge period in vaccinated respect to unvaccinated patients. These findings

Table 1 Main characteristics of the four studies included in the meta-analysis

Authors (year) Country Source of data	Study design Patients included (vaccinated/ unvaccinated) Study period	Age Gender	Outcomes (for vaccinated patients)	remarks
- Kopel E <i>et al.</i> (2014)²⁸ - Israel - Heart Failure Survey in Israel (HFSIS)	- Cohort study 1964 (501/1463) - March–April 2003	74 years (mean) 56% male	- In-hospital 30-day all-cause mortality: - Adjusted HR 0.71; 95% CI 0.42–1.18^a 1-year all-cause mortality: Adjusted HR 0.81; 95% CI 0.66–0.99^a 4-year all-cause mortality: Adjusted HR 0.83; 95% CI 0.73–0.95^a	- All public Israeli hospitals participated - Included hospitalized patients (aged ≥ 50 y) with a diagnosis of AHF. - Evaluation of effects of seasonal vaccination (administered ≤ 12 months before the AHF event) - Neither the vaccine type nor the specific influenza strain administered was reported, and it is unclear whether they matched the circulating seasonal strains. - The study only collected data for 2 months, which may limit its generalization to other seasons or flu outbreaks.
- Wu WC <i>et al.</i> (2014)²⁶ - USA - Hospital Inpatient Quality Reporting (IQR)	- Retrospective study 2616 (2087/429) 2001–2007	70 years (median) 98.2% male	30-day post-discharge all-cause mortality: Adjusted OR 0.49 (95% CI 0.29–0.83)^b 1-year post-discharge all-cause mortality: Adjusted OR 0.74 (95% CI 0.58–0.96)^b	- Sample made up of males (veterans centres), results may not be generalized to women. - Included hospitalized patients with a diagnosis of AHF and discharged alive (patients dying during the index event were excluded) - Evaluation of effects of seasonal vaccination (administered between September and spring) and of vaccination during the AHF episode (but did not distinguish nor analyse these two subgroups) - Neither the vaccine type nor the specific influenza strain administered was reported, and it is unclear whether they matched the circulating seasonal strains. - The use of administrative data may lead to errors in classification or patient characteristics, mainly regarding vaccination or comorbidities. - Took place in a government-operated integrated healthcare system which differs from the private sector
- Bhatt AS <i>et al.</i> (2018)²⁷ - USA - GWTHF-HF registry	- Retrospective study 64 614 (25 873/38 741) - October 2012 to December 2015	75 years (median) 51.5% male	1-year all-cause mortality: Adjusted HR 0.96; 95% CI 0.89–1.03^c 1-year all-cause hospitalization: Adjusted HR 0.97; 95% CI 0.92–1.03^c	- GWTHF-HF voluntary process programme may be biased to include hospitals with a higher likelihood of following evidence-based recommendations - Included hospitalized patients with a diagnosis of AHF and discharged alive (patients dying during the index event were excluded) - Evaluation of effects of seasonal vaccination (administered ≤ 12 months before the AHF event) - Neither the vaccine type nor the specific influenza strain administered was reported, and it is unclear whether they matched the circulating seasonal strains. - Characteristics at a single point in time does not capture all the relevant variables - Included patients with a diagnosis of AHF in the ED, irrespective of whether the patients were hospitalized or sent home after ED care. - Evaluation of effects of seasonal vaccination (administered between October and November) - Neither the vaccine type nor the specific influenza strain administered was reported, and it is unclear whether they matched the circulating seasonal strains.
- Miró O <i>et al.</i> (2023)²⁰ - Spain - EAHFE Registry	- Cohort study 6147 (1139/5008) - January–February 2018 and January–February 2019	84 years (median) 44.4% male	- In-hospital all-cause mortality: Adjusted OR 0.965; 95% CI 0.747–1.246^d 90-day all-cause death:	- Included patients with a diagnosis of AHF in the ED, irrespective of whether the patients were hospitalized or sent home after ED care. - Evaluation of effects of seasonal vaccination (administered between October and November) - Neither the vaccine type nor the specific influenza strain administered was reported, and it is unclear whether they matched the circulating seasonal strains.

Table 1 Continued

Authors (year) Country Source of data	Study design Patients included (vaccinated/ unvaccinated) Study period	Age Gender	Outcomes (for vaccinated patients)	remarks
Miró O et al. (2025) ²¹ Spain EAHFE Registry	Cohort study 4243 (1841/2402) November-December 2022	85 years (median) 43.4% male	- Adjusted HR 0.885; 95% CI 0.750–1.044^d 90-day post-discharge combined adverse events: - Adjusted HR 1.024; 95% CI 0.920–1.140^d 1-year all-cause mortality^e: - Adjusted HR 0.878; 95% CI 0.775–0.994^d - In-hospital all-cause mortality: - Adjusted OR 0.761; 95% CI 0.583–0.992^f 90-day all-cause death: - Adjusted HR 0.831; 95% CI 0.709–0.973^f 1-year all-cause mortality^e - Adjusted HR 0.885; 95% CI 0.785–0.999^f	- Very elderly AHF patients - Included patients with a diagnosis of AHF in the ED, irrespective of whether the patients were hospitalized or sent home after ED care. - Evaluation of effects of seasonal vaccination (administered between September and October) - Neither the vaccine type nor the specific influenza strain administered was reported, and it is unclear whether they matched the circulating seasonal strains. - The post-pandemic period may have an influence on clinical management and higher vaccination rates. - Very elderly AHF patients

^aCovariates used for adjustment were age, gender, new-onset heart failure presentation, heart failure as primary admission diagnosis, NYHA score, left ventricle ejection fraction, obesity, systolic blood pressure, plasma glucose, estimated glomerular filtration rate, total cholesterol, family history of cardiac disease, co-morbidity index (an enumeration of the following comorbidities: hypertension, diabetes mellitus, peripheral vascular disease, chronic obstructive pulmonary disease, past cerebrovascular stroke or myocardial infarction), current or past smoker, prior hospital admissions in past 12 months, marital status, and preadmission drug treatments (aspirin or clopidogrel or warfarin, angiotensin converting enzyme inhibitor or angiotensin receptor blocker, β -blockers, diuretics, calcium channel blockers, amiodarone or other anti-arrhythmics, statins, insulin, oral hypoglycaemics and tranquilizers or antidepressants).

^bCovariates used for adjustment were age; sex; race; hospital days in the last 6 months; number of hospitalizations prior to 6 months; prior HF hospitalization; Elkhäuser risk index; prior myocardial infarction; fiscal year of the assessment; values of haematocrit, mean arterial blood pressure, pulse, and creatinine clearance; and clustering within hospitals.

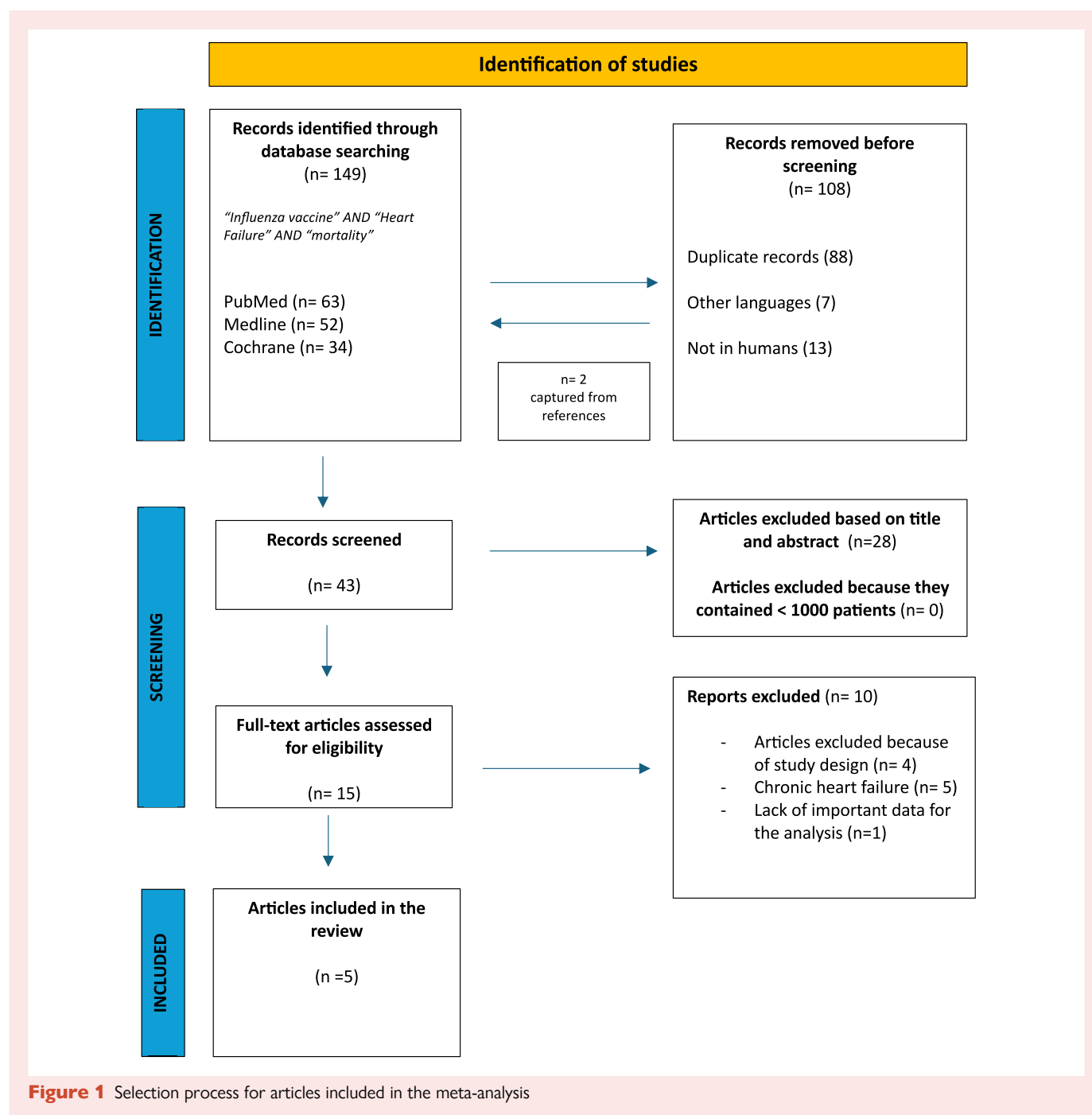
^cCovariates used for adjustment were age, sex, race, insurance, comorbidities (atrial fibrillation/flutter, chronic obstructive pulmonary disease, diabetes mellitus, dyslipidaemia, hypertension, peripheral vein disease, stroke, anaemia, pacemaker, dialysis, renal insufficiency, depression, valvular disease, implanted cardiac defibrillator, cardiac resynchronization therapy pacemaker, ischaemic heart disease, smoker, and left ventricular dysfunction) and hospital characteristics (teaching hospital, number of beds, region, rural vs urban, percutaneous coronary intervention, surgical, interventional and heart transplantation capability, number of cardiac intensive care beds, adult cardiology service, mobile health service, established medical home, has adult diagnostic catheterization capability, and has adult interventional catheterization capability).

^dCovariates used for adjustment were age, sex, season of vaccination, baseline status (assessed by NYHA class and Barthel index as baseline), infection as trigger of decompensation and comorbidities (liver cirrhosis, heart valve disease, chronic kidney disease, peripheral artery disease, active neoplasia, diabetes mellitus, dementia, and chronic obstructive pulmonary disease).

^eThis hazard ratio was not published in the original paper and was reported by authors on specific demand for the present meta-analysis.

^fCovariates used for adjustment were age, sex, comorbidities (hypertension, diabetes mellitus, coronary artery disease, previous AHF episodes, atrial fibrillation, chronic kidney disease, heart valve disease, chronic obstructive pulmonary disease, dementia, neoplasia, cerebrovascular disease), baseline functional situation of the patient (Barthel Index, NYHA class, left ventricular dysfunction), chronic treatments (diuretics, beta-blockers, renin-angiotensin system inhibitors, mineral corticosteroid-receptor antagonists, neprilysin-inhibitors, sodium-glucose cotransporter 2 inhibitors), COVID-19 vaccination and vitals at emergency department arrival (systolic blood pressure, heart rate, pulse-oximetry), signs of low cardiac output, identification of infection as the precipitating factor and laboratory results (haemoglobin, creatinine, sodium, potassium, N-terminal pro-brain natriuretic peptide) at emergency department arrival.

OR, odds ratio; HR, hazard ratio; CI, confidence interval; AHF, acute heart failure; ED, emergency department; IQR, interquartile range.



suggest that the relationship between influenza vaccination and decreased mortality is not only limited to the long-term, but that there are already measurable associations during hospitalization and within the first 3 months following an AHF episode.

Finally, some outcomes were reported only once. Wu *et al.*²⁶ reported an adjusted OR of 0.49 (95% CI 0.29–0.83) for 30-day mortality, which is a significant reduction that is in line with the results presented previously. Conversely, Bhatt *et al.*²⁷ reported a non-significant relationship between influenza vaccination and 1-year all-cause hospitalization (adjusted HR 0.97; 95% CI 0.92–1.03), and Miró *et al.*²⁰ reported no relationship with 90-day post-discharge combined adverse events (which included all-cause death and hospitalization and ED revisit due

to AHF; adjusted HR 1.024; 95% CI 0.920–1.140). These results suggest that the association of vaccination of patients with AHF with reconsultation and hospitalization rates is not relevant.

Discussion

To our knowledge, this meta-analysis is the first to specifically address the effect of influenza vaccination on patients experiencing an episode of AHF. Previous literature has largely focused on stable chronic HF populations, with limited insight into acute decompensation settings. While it is well-known that influenza vaccination prevents

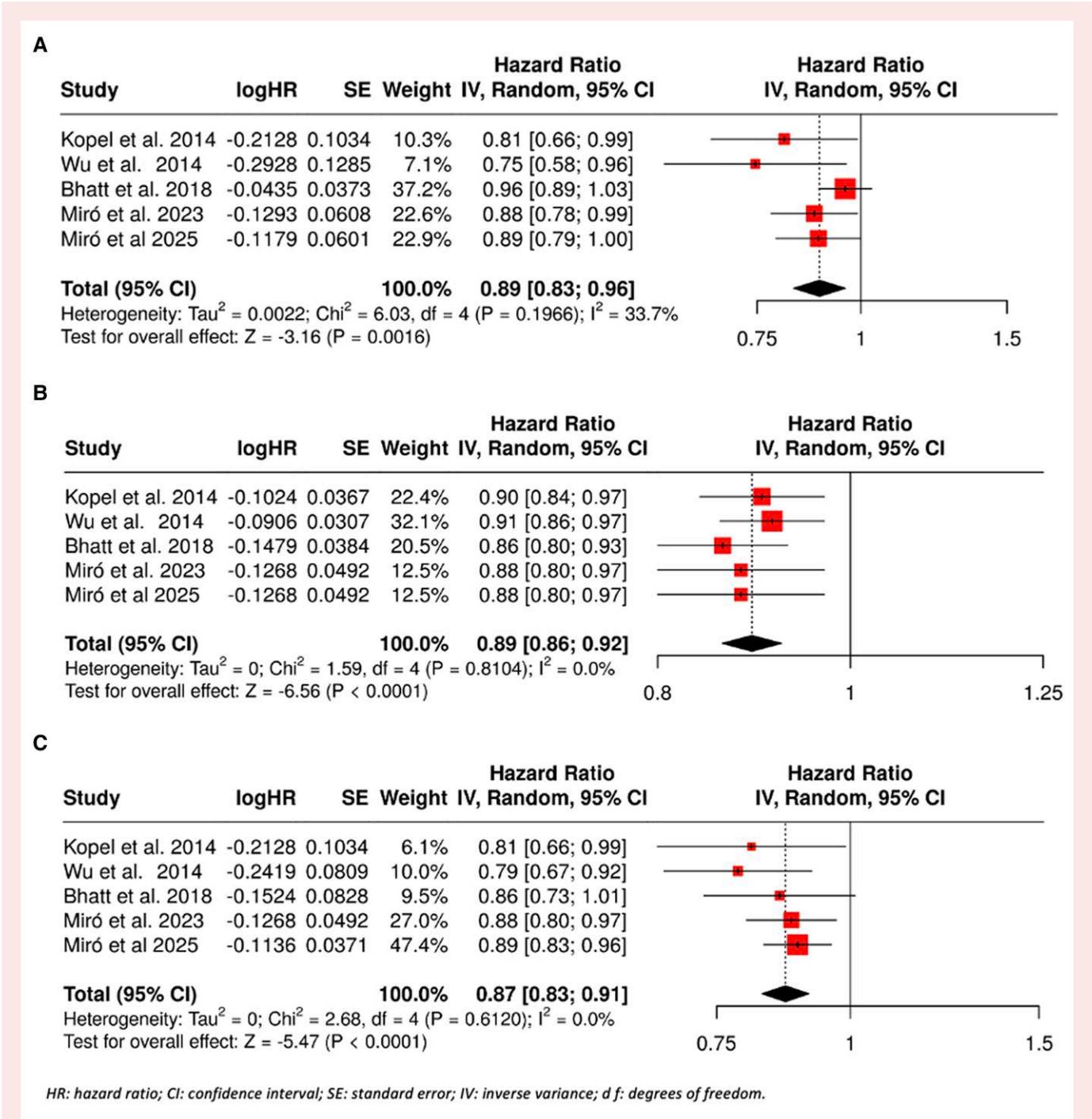


Figure 2 Forest plots of the main analysis of association between influenza vaccination and the primary outcome (1-year all-cause mortality). (A) shows the meta-analysis including the five studies; (B) represents the sensitivity after excluding each study once and repeating the pooled estimations of the association between influenza vaccination and all-cause mortality using the remaining studies; and (C) depicts the results of the cumulative meta-analysis over time in the sequence the studies were published. Squares indicate study-specific hazard ratio estimates (the size of the square reflects the study-specific statistical weight); horizontal lines indicate the 95% confidence interval; black diamond indicates the pooled hazard ratio. HR, hazard ratio; CI, confidence interval; SE, standard error; IV, inverse variance; df, degrees of freedom

decompensations due to influenza infection in patients with chronic HF, there is little known about the potential effects of vaccination in patients with AHF, regardless of the cause of decompensation. Our findings are based on five large observational cohort studies and provide

evidence that influenza vaccination is associated with a modest, albeit statistically significant, reduction in all-cause mortality at multiple time points—including in-hospital (probably), 90-day, and 1-year outcomes—in the specific setting of patients with AHF.

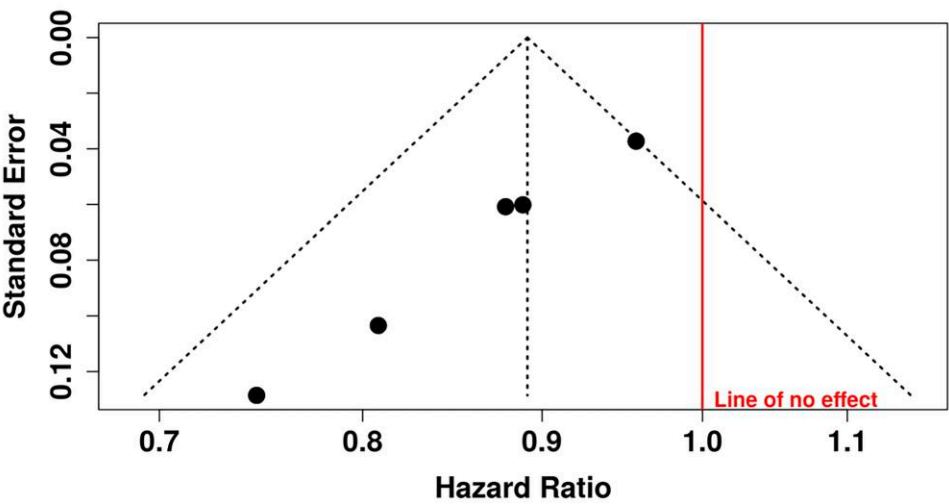


Figure 3 Funnel plot for studies included in the primary outcome meta-analysis (1-year all-cause mortality)

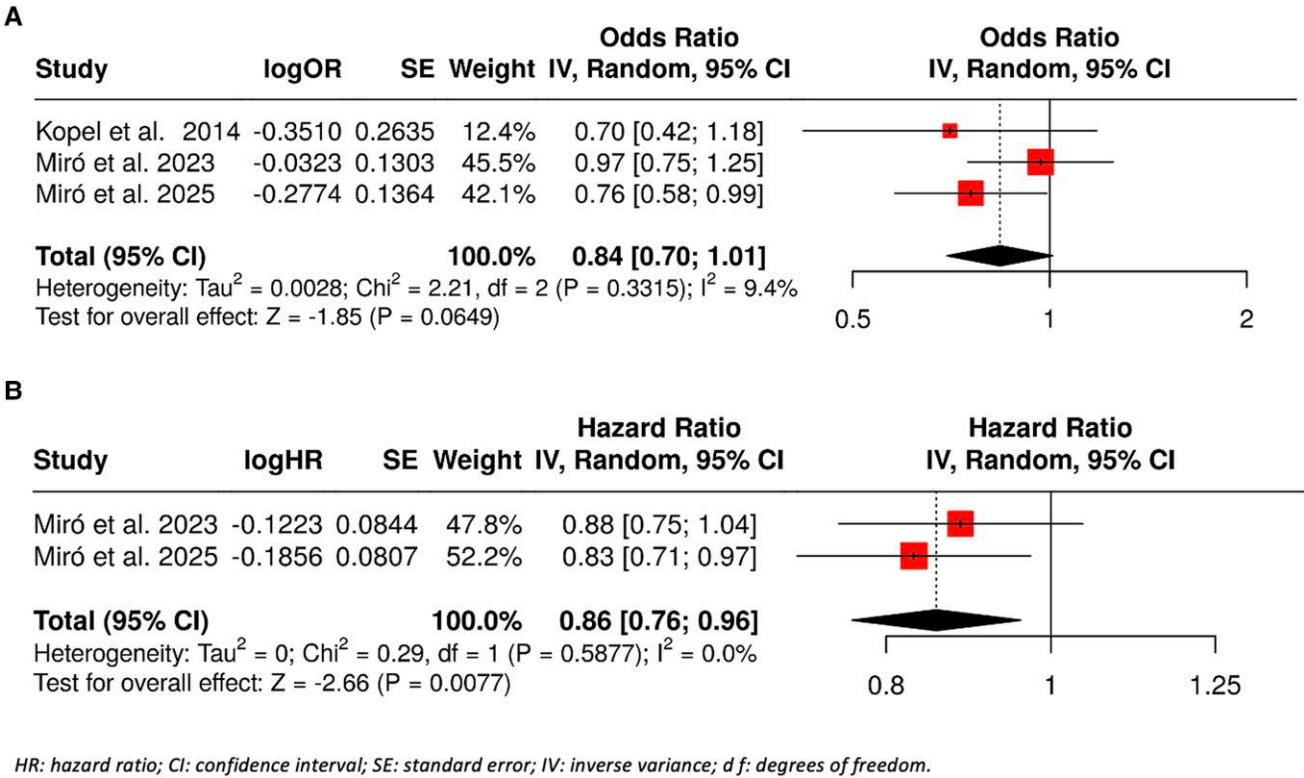


Figure 4 Forest plots of the main analysis of association between influenza vaccination and the secondary outcomes [(A): in-hospital all-cause mortality; (B): 90-day all-cause mortality]. HR, hazard ratio; CI, confidence interval; SE, standard error; IV, inverse variance; df, degrees of freedom

The pooled HR of 0.89 (95% CI: 0.83–0.96) for 1-year all-cause mortality suggests a relative decrease of 11%, which was consistent across studies, despite variability in geographical context, health systems, vaccination coverage, and baseline patient characteristics. Importantly,

heterogeneity was low ($I^2 = 33.7\%$), and sensitivity analyses confirmed the robustness of the results, reinforcing the consistency of the associations observed. These results were built upon the evidence reported in previous observational studies in chronic HF populations. For example,

Modin et al. found a significant benefit of influenza vaccination related to mortality in HF patients, with the strongest effect observed in those aged ≥ 65 years and during influenza seasons.^{28,29} Similarly, Gotsman et al.¹³ demonstrated that vaccinated HF patients had a significantly lower all-cause mortality and cardiovascular-related hospitalizations compared with unvaccinated individuals. Although these studies did not focus on AHF episodes *per se*, they provide biological plausibility for our findings.

Remarkably, after the inclusion phase of this meta-analysis had closed (on 30 June 2025), the results of the PANDA-II randomized clinical trial were published.³⁰ This trial was conducted in 164 hospitals across China, which were randomized (1:1) in batches before each winter enrolment period to either the intervention or control group. The intervention consisted of offering free influenza vaccination during hospitalization to all patients admitted with AHF, while hospitals in the control group continued routine care, whereby patients were advised at discharge that they could receive influenza vaccination at a community healthcare centre. Over three consecutive winter seasons, 3570 patients were enrolled in the intervention group [3368 (94%) of whom were vaccinated during hospitalization] and 4201 in the control group [21 (0.5%) of whom received vaccination]. The 1-year post-discharge all-cause mortality or hospital readmission for any cause was the composite primary outcome and occurred in 41.2% of the intervention group and 47.0% of the control group, corresponding to an OR of 0.83 (95% CI 0.72–0.97) in vaccinated patients. These results demonstrate that influenza vaccination administered during hospitalization due to AHF can improve survival and reduce the likelihood of hospital readmission over the subsequent 12 months. We believe that the non-experimental evidence summarized in the present meta-analysis is consistent with the findings reported in the PANDA-II trial. Therefore, in line with the proposal of the PANDA-II investigators, we consider that integrating influenza vaccination into inpatient care should be regarded as a strategy to improve outcomes in patients with HF in general, and in those hospitalized for AHF in particular.

The protective role of vaccination may extend beyond direct prevention of influenza infection. On one hand, influenza infection in vaccinated patients can be less severe than in those who are unvaccinated, as marked reductions in the length of hospital stay, intensive care unit admission and in-hospital death have been demonstrated in adults vaccinated for influenza who were hospitalized with laboratory-confirmed influenza.³¹ In this regard, although the presence of active influenza infection was not reported in any of the studies included in the meta-analysis, it is reasonable to assume that some patients included in such studies (in unvaccinated but also in vaccinated groups) may have had influenza infection. It is well-known that the efficacy of influenza vaccination is not generalized in vaccinated populations, with a previous extensive meta-analysis showing a pooled efficacy of 59%.³² Despite this fact, our data suggest that vaccine provided direct protection to vaccinated individuals with AHF, especially in terms of long-term mortality. Moreover, respiratory infections are well-established triggers for HF decompensation.^{10,33} Prevention of these infections that may arise after a flu episode may reduce the burden on already compromised cardiac function. Additionally, several studies suggest that influenza infection may contribute to endothelial dysfunction, plaque instability, and systemic inflammation, all of which could exacerbate the pathophysiology of HF weeks after the influenza episode, making decompensations in unvaccinated patients more frequent and severe.³⁴ Notably, our findings also suggest that the benefits of influenza vaccination extend into the early post-discharge period, which supports the concept of the 'vulnerable phase' after AHF, during which patients face a heightened risk of adverse outcomes^{7,9} and implies that vaccination may play a role in mitigating early rehospitalization or death.

On the other hand, influenza is linked not only to infection-related mortality but also to acute coronary events and other adverse cardiovascular outcomes. In a recent analysis, Davidson et al. reported that influenza vaccination significantly reduced the risk of acute

cardiovascular events, with larger reductions in relative risk shortly after vaccination (28% at 15–28 days) than later (16% at 91–120 days).¹⁷ Among specific events, the reduction was more pronounced for myocardial infarction (40% at 15–28 days) than for other cardiovascular endpoints. The mechanisms underlying these benefits remain uncertain. One possibility is the prevention of influenza infection itself, which can precipitate cardiovascular events. Another is vaccine-mediated pleiotropic effects—such as cross-reactivity between viral antigens and atherosclerotic plaque components and non-specific immunomodulatory actions—that may reduce cardiovascular risk even in the absence of circulating influenza virus.³⁵ Regardless of the precise mechanism(s), vaccination-related prevention of acute coronary syndromes—and consequent reductions in mortality—likely contributed, in part, to the associations observed in the present meta-analysis.

Nonetheless, the analysis also revealed that influenza vaccination does not significantly reduce rehospitalizations or ED revisits, particularly in the 90-day post-discharge window, as seen in the studies by Bhatt et al.²⁷ and Miró et al.²⁰ This suggests that while mortality is impacted, the broader burden of healthcare utilization may remain unchanged, possibly due to non-influenza-related factors, such as frailty, polypharmacy, or socioeconomic determinants, that vaccination does not influence directly. On the other hand, both ED revisits and hospitalizations are less severe outcomes than mortality, and they also depend, to some extent, on non-medical patient-related conditions (i.e. familiar and social support or proximity to hospital) and healthcare system-related factors (including ease of patient hospitalization or the availability of alternative facilities to treat patients with HF; which can differ from country to country and even from hospital to hospital in the same country).^{36–38} These findings related to ED revisits and rehospitalizations should be interpreted with caution. They are based on single observational studies and require re-evaluation as additional evidence becomes available. Moreover, length of stay for the index hospitalization, the number of rehospitalizations or specific causes leading to rehospitalization, are not reported. Moreover, no competing-risks analyses were performed to account for death as a competing event, rendering inferences about these outcomes particularly uncertain. Nonetheless, the findings of the present meta-analysis are consistent with those observed in the PANDA-II randomized clinical trial,³⁰ in which the reduction in rehospitalization among vaccinated patients over both 6- and 12-month follow-up periods was not statistically significant after adjustment and was smaller than the reduction observed for mortality (OR 0.83 vs 0.74 at 6 months, and 0.83 vs 0.76 at 12 months).

A critical observation is the relatively low vaccination rate among AHF patients in some cohorts, especially in Southern Europe. For instance, the Miro et al. study²⁰ reported that only 19% of AHF patients had received the influenza vaccine,²¹ despite the recommendations of the guidelines of the European Society of Cardiology strongly advocating seasonal influenza vaccination in all HF patients as part of standard preventive care.³⁹ Our findings also underscore an urgent public health opportunity: targeted strategies are needed to improve vaccination coverage in HF populations. Barriers may include vaccine hesitancy, lack of awareness among clinicians, and under prioritization during acute hospital care.⁴⁰ Given the evidence presented here, integration of vaccination into chronic HF management pathways or even into discharge planning after an AHF episode (one of the studies included in this meta-analysis included patients vaccinated during the index AHF episode in the group of vaccinated patients, as well as patients included in the PANDA-II trial) could be a cost-effective and low-risk intervention to improve outcomes.

Limitations

First, all the studies included were observational studies and, despite all having adjusted the risk for different covariates, some remaining confounders may remain. Vaccinated patients may inherently exhibit better

adherence to treatment, greater healthcare engagement, and fewer social vulnerabilities. Second, the number of eligible studies was small ($n = 5$), which restricted the ability to perform comprehensive subgroup analyses and precluded formal statistical assessment of publication bias such as Egger's test. Although visual inspection of the funnel plot did not suggest substantial asymmetry, the small number of studies limits the robustness of this analysis. Third, vaccination reporting was heterogeneous and incomplete. Although all the studies defined vaccination as receipt of the seasonal vaccine, the interval between vaccination and the index AHF episode ranged from several weeks to as long as 12 months; in one study, even doses administered during the index hospitalization also qualified as vaccination. No study reported the vaccine type (e.g. trivalent vs quadrivalent; standard vs high-dose) or assessed antigenic match with circulating strains. This heterogeneity limited our ability to evaluate differential effects by timing, formulation, or strain match. Fourth, the clinical and demographic characteristics of the study populations were heterogeneous. Notably, one study included only male veterans, while others focused on very elderly populations treated in EDs in Europe. This variability may affect the external validity and generalizability of our findings to the broader HF populations, especially among women and younger patients. Fifth, the reduction in adverse outcomes at 1 year among patients with AHF who received influenza vaccination cannot be fully attributed to the effect of the vaccine itself. Length of stay and the number of adverse events during hospitalization—including cardiovascular or all-cause mortality—would likely provide a better representation of outcomes directly related to the AHF episode and the ability of the vaccine to prevent them, as would adverse events occurring during the first 3 months after discharge (the so-called vulnerable period). However, these outcomes were less frequently reported in the studies included in the meta-analysis, and their results were not as consistent as those observed for 1-year mortality. Lastly, some relevant secondary outcomes, such as rehospitalization or ED revisits, were reported in only one or two studies and showed no significant association with vaccination. Additionally, no studies investigated the length of hospitalization, which can also be impacted by vaccination. Therefore, all these findings should be interpreted with caution, as they may reflect differences in healthcare systems, admission thresholds, or patient-level socioeconomic factors rather than a true lack of vaccine effect.

Conclusion

Influenza vaccination seems to be associated with lower all-cause mortality in patients with AHF, although all the studies included were observational, and thus, residual confounding cannot be ruled out. Nonetheless, the consistency of our findings with previous literature on both chronic HF and broader cardiovascular populations, as well as the very recently published results of the PANDA-II trial, reinforce the reliability of our results. Accordingly, efforts must be redoubled to ensure that all patients with HF receive an annual flu vaccination. Moreover, observational findings summarized in the present meta-analysis, now complemented by the randomized PANDA-II trial, collectively support the routine offer of influenza vaccination to patients hospitalized for AHF.

Declarations

Disclosure of Interest

All authors declare no disclosure of interest for this contribution.

Data Availability

The data is not open to the public, but if there is a reasonable request the authors might share it.

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